

Algorithmic Equivalence with Extension Types and Disjunction Eliminations

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Everett (Trey) Plante III, Robert Rose, and Dan Licata

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A lot of type theories...

- $\triangle$ Simplicial type theories [Riehl, Shulman], [Bucholtz, Weinberger], [Gratzer, Weinberger, Buchholtz], [Pugh, Sterling]
- $\square$ Cubical type theories [CCHM], [AFH], [Orton, Pitts], [ABCFHL], [Weaver, Licata]
- Φ Type theories with propositional layers [Sterling, Harper], [Gratzer, Sterling, Angiuli, Coquand, Birkedal]

Various implementations in rzk, Cubical Agda, redtt, cooltt, cubicaltt, Arend, ...

Terms ($\mathcal{T}$)

$\mathbb{I} \quad \mathbb{Z} \quad p \quad \neg i \quad i \vee j \quad i \wedge j \quad \dots$

Formulas ($\mathcal{F}$)

$\phi \wedge \psi \quad \phi \vee \psi \quad \phi \rightarrow \psi \quad (i = 0) \quad (i = j) \quad \dots$

Judgments

$$\Xi, i : \mathcal{T} \mid \Phi, \alpha \mid \Delta, x : \tau \vdash J$$

Judgmental Entailment

$$\Xi \mid \Phi \mid \Delta \vdash \Psi$$

I will write judgments as $\Gamma \vdash J$ and entailments as $\Gamma \vdash \psi$.

$$\frac{\Gamma \vdash \tau \quad \Gamma, i : \mathbb{I} \vdash M : \tau}{\Gamma \vdash \langle i \rangle M : \text{Path}_\tau(M_0, M_1)} \quad \frac{\Gamma \vdash N : \text{Path}_\tau(M_0, M_1)}{\Gamma \vdash N \ 0 \equiv M_0 : \tau}$$

$$\frac{\Gamma \vdash \tau \quad \Gamma, i : \mathbb{Z} \vdash M : \tau}{\Gamma \vdash \langle i \rangle M : \text{Hom}_\tau(M_0, M_1)}$$

Disjunctions of Formulas

$$\frac{\Gamma, (i = 0) \vdash M_0 : \tau \quad \Gamma, (i = 1) \vdash M_1 : \tau}{\Gamma, (i = 0) \vee (i = 1) \vdash \text{case}((i = 0).M_0, (i = 1).M_1) : \tau}$$

- Systems [CCHM], [Sterling, Harper]
- Join of compatible partial elements [Orton, Pitts]
- Cases [ABCFHL], [AFH]
- Disjunction Elimination [Riehl, Shulman]

$$\frac{\Gamma \vdash P : \text{Path}_\tau(M_0, M_1) \quad \Gamma \vdash Q : \text{Path}_\tau(M_1, M_2)}{\Gamma \vdash \langle i \rangle \text{comp}^j(\tau[(i = 0) \mapsto M_0, (i = 1) \mapsto Q@j](P@i) : \text{Path}_\tau(M_0, M_2)}$$

Extension

$$\frac{\Gamma \vdash \tau \quad \Gamma \vdash \psi : \mathcal{F} \quad \Gamma, \psi \vdash t : \tau}{\Gamma \vdash \{\tau \mid \psi \mapsto t\}} \quad \frac{\Gamma \vdash M : \tau \quad \frac{\Gamma, \psi \vdash t : \tau \quad \Gamma, \psi \vdash M \equiv t : \tau}{\Gamma \vdash \text{in}_{\psi}^t M : \{\tau \mid \psi \mapsto t\}}}{\Gamma \vdash \text{in}_{\psi}^t M : \{\tau \mid \psi \mapsto t\}}^*$$

- Extension types [Riehl, Shulman], [Gratzer et. al.]
- Extent types/signatures [Sterling], [Sterling, Harper]
- Extension of cofibrant partial elements [Orton, Pitts]

*This is a particular flavor of extension types.

Reconstructing Paths and Morphisms

$$\frac{\Gamma, i : \mathbb{I}, (i = 0) \vdash M_0 : \tau \quad \Gamma, i : \mathbb{I}, (i = 1) \vdash M_1 : \tau}{\Gamma, i : \mathbb{I}, (i = 0) \vee (i = 1) \vdash \text{case}((i = 0).M_0, (i = 1).M_1) : \tau}$$
$$\frac{\Gamma, i : \mathbb{I} \vdash M : \tau \quad \Gamma, i : \mathbb{I}, (i = 0) \vee (i = 1) \vdash M \equiv \text{case}((i = 0).M_0, (i = 1).M_1) : \tau}{\Gamma, i : \mathbb{I} \vdash \text{in } M : \{\tau \mid (i = 0) \vee (i = 1) \mapsto \text{case}((i = 0).M_0, (i = 1).M_1)\}}$$
$$\frac{\Gamma, i : \mathbb{I} \vdash \text{in } M : \{\tau \mid (i = 0) \vee (i = 1) \mapsto \text{case}((i = 0).M_0, (i = 1).M_1)\}}{\Gamma \vdash \lambda i : \mathbb{I}. \text{in } M : (i : \mathbb{I}) \rightarrow \{\tau \mid (i = 0) \vee (i = 1) \mapsto \text{case}((i = 0).M_0, (i = 1).M_1)\}}$$

Paths/Morphisms are implemented in STT [Riehl, Shulman] and cooltt with extension types.

Extension Types

$$\frac{\Gamma \vdash \tau \quad \Gamma \vdash \phi : \mathcal{F} \quad \Gamma, \phi \vdash t : \tau}{\Gamma \vdash \{\tau \mid \phi \mapsto t\}}$$

$$\frac{\Gamma \vdash M : \tau \quad \Gamma, \psi \vdash t : \tau \quad \Gamma, \psi \vdash M \equiv t : \tau}{\Gamma \vdash \text{in}_{\psi}^t M : \{\tau \mid \psi \mapsto t\}}$$

$$\frac{\Gamma \vdash \text{in}_{\psi}^t O : \{\tau \mid \psi \mapsto M\}}{\Gamma \vdash \text{out}_{\psi}^t \text{in}_{\psi}^t O \equiv O : \tau} \quad \frac{\Gamma \vdash M : \{\tau \mid \psi \mapsto t\} \quad \Gamma \vdash \psi}{\Gamma \vdash \text{out}_{\psi}^t M \equiv t : \tau}$$

$$\frac{\Gamma \vdash M : \{\tau \mid \psi \mapsto t\}}{\Gamma \vdash \text{in}_{\psi}^t \text{out}_{\psi}^t M \equiv M : \{\tau \mid \psi \mapsto t\}}$$

Types do not have to be fibrant.

Disjunction Elimination

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \Gamma, \alpha_1 \vdash M_1 : \tau \quad \Gamma, \alpha_2 \vdash M_2 : \tau \quad \Gamma, \alpha_1, \alpha_2 \vdash M_1 \equiv M_2 : \tau}{\Gamma \vdash \text{case}(\alpha_1.M_1, \alpha_2.M_2) : \tau}$$

$$\frac{\Gamma \vdash \alpha_i}{\Gamma \vdash \text{case}(\alpha_1.M_1, \alpha_2.M_2) \equiv M_i : \tau}$$

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \Gamma \vdash M : \tau}{\Gamma \vdash M \equiv \text{case}(\alpha_1.M, \alpha_2.M) : \tau}$$

$$\frac{\Gamma \vdash \alpha_1 \leftrightarrow \beta_1 \quad \Gamma \vdash \alpha_2 \leftrightarrow \beta_2 \quad \Gamma, \alpha_1 \vdash M_1 \equiv N_1 : \tau \quad \Gamma, \beta_1 \vdash M_2 \equiv N_2 : \tau}{\Gamma \vdash \text{case}(\alpha_1.M_1, \alpha_2.M_2) \equiv \text{case}(\beta_1.N_1, \beta_2.N_2) : \tau}$$

Underlying Structure and Pattern

Type theories with decidable proof-irrelevant logics, extension types, and disjunction elimination.

- Add an underlying term structure to a type theory $\mathcal{T}$ ($\mathbb{I}$, **2**, some structured interval usually)
- Define a formula language $\mathcal{F}$ generated from equalities of terms and close them under $\vee$, $\wedge$ (maybe other stuff too)
- Design new feature you wanted/Formalize desired mathematics.
- Wish you had an implementation ... :(

Example: bicubical directed type theory

$$\begin{array}{c} \overline{\vdash 0 : \mathbb{I}} \quad \overline{\vdash 1 : \mathbb{I}} \quad \overline{\vdash x_i : \mathbb{I}} \\[10pt] \frac{\Psi \vdash p : \mathbb{I} \quad \Psi \vdash q : \mathbb{I}}{\Psi \vdash p \vee q : \mathbb{I}} \quad \frac{\Psi \vdash p : \mathbb{I} \quad \Psi \vdash q : \mathbb{I}}{\Psi \vdash p \wedge q : \mathbb{I}} \\[10pt] \overline{\vdash \perp : \mathbf{cof}} \quad \overline{\vdash \top : \mathbf{cof}} \quad \frac{\Psi \vdash t : \mathbb{I} \quad \Psi \vdash s : \mathbb{I}}{\Psi \vdash (t = s) : \mathbf{cof}} \quad \overline{\vdash \alpha_i : \mathbf{cof}} \\[10pt] \frac{\Psi \vdash \alpha : \mathbf{cof} \quad \Psi \vdash \beta : \mathbf{cof}}{\Psi \vdash \alpha \vee \beta : \mathbf{cof}} \quad \frac{\Psi \vdash \alpha : \mathbf{cof} \quad \Psi \vdash \beta : \mathbf{cof}}{\Psi \vdash \alpha \wedge \beta : \mathbf{cof}} \\[10pt] \frac{\Psi \vdash \phi : \mathbf{cof}}{\Psi \vdash \forall x_i. \phi : \mathbf{cof}} \end{array}$$

- Can we solve $\phi \vdash \psi$?

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- Can we typecheck a $\lambda^{\Box}$ with extension types and disjunction eliminations?

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- Can we typecheck a $\lambda^{\Box}$ with extension types and disjunction eliminations?

Logic-agnostic

Can we typecheck $\lambda^{\mathcal{L}}$ with extension types and disjunction eliminations?

Implementation Framework for $\lambda^{\mathcal{L}}$

Assume we have an oracle for $\phi \vdash \psi$ [Rose, Licata] and build a type checker for $\lambda^{\mathcal{L}}$.

$$\frac{\Gamma \vdash \tau_1 \equiv \tau_2 \quad \Gamma \vdash M : \tau_1}{\Gamma \vdash M : \tau_2}$$

$\lambda^{\mathcal{L}}$ has dependent functions, no universes yet.

$$\Gamma \triangleright \tau_1 \Leftrightarrow \tau_2 \text{ iff } \Gamma \vdash \tau_1 \equiv \tau_2$$

$$\Gamma \triangleright M \Leftrightarrow N : \tau \text{ iff } \Gamma \vdash M \equiv N : \tau$$

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \Gamma \vdash M : \tau}{\Gamma \vdash M \equiv \text{case}(\alpha_1.M, \alpha_2.M) : \tau}$$

η + congruence = disjunction extensionality

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \begin{array}{l} \Gamma, \alpha_1 \vdash M \equiv N : \tau \\ \Gamma, \alpha_2 \vdash M \equiv N : \tau \end{array}}{\Gamma \vdash M \equiv N : \tau} \text{ disj. ext.}$$

$$\alpha_1 \vee \alpha_2 \vdash M \equiv^\eta \text{case}(\alpha_1.M, \alpha_2.M) \equiv^{\text{cong}} \text{case}(\alpha_1.N, \alpha_2.N) \equiv^\eta N : \tau$$

Disjunction Extensionality in Practice

$\text{loop} : (x : \mathbb{I}) \rightarrow \{S^1 \mid (x = 0) \vee (x = 1) \mapsto \text{case}((x = 0).\text{base}, (x = 1).\text{base})\}$
 $\text{loop } x : \{S^1 \mid (x = 0) \vee (x = 1) \mapsto \text{case}((x = 0).\text{base}, (x = 1).\text{base})\}$
 $\text{out loop } x : S^1$

$(x = 0) \vee (x = 1) \vdash (x = 0) \vee (x = 1)$	$(x = 0) \vdash \text{out loop } x \equiv \text{base} : S^1$
	$(x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1$
$(x = 0) \vee (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1$	

Previous Approaches

- Cubical Agda implements DeMorgan cubes and supports a restricted form of disjunction eliminations in which a user can only write formulas of the form $r = 1$ where $r := \bigvee_i \bigwedge \alpha_i$ where each α_i is atomic [Vezzosi, Mörtberg, Abel]
- cooltt implements Cartesian cubes and supports disjunction eliminations via an underlying transformation of the cofibration language (left inversion) [redPRL].
- rzk implements a simplified tope layer of STT and supports disjunction eliminations via an experimental prover for the tope layer [Kudasov].

We want to implement η with more expressive logics (bicubical).
Type-checking and logic can be separate components.

Type-directed Equivalence Algorithm

$$\begin{array}{ll} \downarrow & \text{many-step weak-head reduction } (\beta\text{-reduction}) \\ \Gamma \triangleright M \Leftrightarrow N : \tau & \text{if } M \downarrow M', N \downarrow N', \Gamma \triangleright M' \Leftrightarrow N' \uparrow \tau', \\ & \text{and } \tau \text{ is neutral} \\ \Gamma \triangleright M \Leftrightarrow N : (x : \tau_1) \rightarrow \tau_2(x) & \text{if } \Gamma, x : \tau_1 \triangleright M \ x \Leftrightarrow N \ x : \tau_2(x) \\ \\ \Gamma \triangleright x \Leftrightarrow x \uparrow \Gamma(x) & \\ \Gamma \triangleright pM \Leftrightarrow qN \uparrow \tau_2[M/x] & \text{if } \Gamma \triangleright p \Leftrightarrow q \uparrow (x : \tau_1) \rightarrow \tau_2(x) \\ & \text{and } \Gamma \triangleright M \Leftrightarrow N : \tau_1 \end{array}$$

[Coquand], [Harper, Stone], [Harper, Pfenning], [Abel, Scherer], [Abel, Coquand, Mannaa], [Abel, Öhman, Vezzosi]

$$\Gamma \vdash \text{out}_{\psi}^t(\text{in}_{\psi}^t(M)) \rightarrow M$$

$$\Gamma \vdash \text{out}_{\psi}^t(M) \rightarrow t \text{ if } \Gamma \vdash \psi$$

$$\Gamma \triangleright M \Leftrightarrow N : \{\tau \mid \psi \mapsto t\} \text{ if } \Gamma \triangleright \text{out}_{\psi}^t(M) \Leftrightarrow \text{out}_{\psi}^t(N) : \tau$$

[Harper, Stone]

Disjunction Extensionality

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \begin{array}{l} \Gamma, \alpha_1 \vdash M \equiv N : \tau \\ \Gamma, \alpha_2 \vdash M \equiv N : \tau \end{array}}{\Gamma \vdash M \equiv N : \tau}$$

$\rightsquigarrow$

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \begin{array}{l} \Gamma, \alpha_1 \vdash \text{case}(\alpha_1.M, \alpha_2.M) \equiv^\beta M : \tau \\ \Gamma, \alpha_2 \vdash \text{case}(\alpha_1.M, \alpha_2.M) \equiv^\beta M : \tau \end{array}}{\Gamma \vdash \text{case}(\alpha_1.M, \alpha_2.M) \equiv M : \tau} \eta$$

Naive Solution

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \begin{array}{l} \Gamma, \alpha_1 \triangleright M \Leftrightarrow N : \tau \\ \Gamma, \alpha_2 \triangleright M \Leftrightarrow N : \tau \end{array}}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$

Generating disjunctions

Given $\Gamma \vdash M \equiv N : \tau$, what are the disjuncts α_i such that $\Gamma, \alpha_i \vdash M \equiv N : \tau$ and $\Gamma \vdash \bigvee_i \alpha_i$?

Generating disjunctions

Given $\Gamma \vdash M \equiv N : \tau$, what are the disjuncts α_i such that $\Gamma, \alpha_i \vdash M \equiv N : \tau$ and $\Gamma \vdash \bigvee_i \alpha_i$?

$$\frac{\begin{array}{l} (x = 0) \vee (x = 1) \vdash \bigvee_i ? \quad ? \vdash \text{out loop } x \equiv \text{base} : S^1 \\ \quad \quad \quad ? \vdash \text{out loop } x \equiv \text{base} : S^1 \end{array}}{(x = 0) \vee (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1}$$

Generating disjunctions

Given $\Gamma \vdash M \equiv N : \tau$, what are the disjuncts α_i such that $\Gamma, \alpha_i \vdash M \equiv N : \tau$ and $\Gamma \vdash \bigvee_i \alpha_i$?

$$\frac{(x = 0) \vee (x = 1) \vdash (x = 0) \vee (x = 1) \quad \begin{array}{l} (x = 0) \vdash \text{out loop } x \equiv \text{base} : S^1 \\ (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1 \end{array}}{(x = 0) \vee (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1}$$

Generating disjunctions

$$\frac{\begin{array}{l} ((x_1 = 0) \vee (x_1 = 1)) \wedge (x_2 = 0) \vdash (x_1 = 0) \vee (x_1 = 1) \\ (x_1 = 0) \wedge (x_2 = 0) \vdash \text{out loop } x_1 \equiv \text{out loop } x_2 : S^1 \\ (x_1 = 1) \wedge (x_2 = 0) \vdash \text{out loop } x_1 \equiv \text{out loop } x_2 : S^1 \end{array}}{((x_1 = 0) \vee (x_1 = 1)) \wedge (x_2 = 0) \vdash \text{out loop } x_1 \equiv \text{out loop } x_2 : S^1}$$

Generating disjunctions

$\text{loop } x : \{S^1 \mid (x = 0) \vee (x = 1) \mapsto \text{case}((x = 0).\text{base}, (x = 1).\text{base})\}$

$$\frac{(x = 0) \vee (x = 1) \vdash (x = 0) \vee (x = 1) \quad \begin{array}{l} (x = 0) \vdash \text{out loop } x \equiv \text{base} : S^1 \\ (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1 \end{array}}{(x = 0) \vee (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1}$$

Generating disjunctions

$\text{loop } x : \{S^1 \mid (x = 0) \vee (x = 1) \mapsto \text{case}((x = 0).\text{base}, (x = 1).\text{base})\}$

$$\frac{(x = 0) \vee (x = 1) \vdash (x = 0) \vee (x = 1) \quad \begin{array}{l} (x = 0) \vdash \text{out loop } x \equiv \text{base} : S^1 \\ (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1 \end{array}}{(x = 0) \vee (x = 1) \vdash \text{out loop } x \equiv \text{base} : S^1}$$

Formula Guarded Reduction

$$\begin{aligned} & \text{case}(\mu_1.M_1, \mu_2.M_2) \rightarrow (M_1, \mu_1), (M_2, \mu_2) \\ \text{out}_{\psi}^t(M) & \rightarrow (t, \psi), (\text{out}_{\psi}^t(M_1), \mu_1), \dots, (\text{out}_{\psi}^t(M_l), \mu_l) \quad \text{if } M \rightarrow \overrightarrow{M}, \mu \\ & (\lambda x : \tau. M) N \rightarrow (M[N/x], \top) \\ M N & \downarrow (M_1 N, \mu_1), \dots, (M_l N, \mu_l) \quad \text{if } M \rightarrow \overrightarrow{M}, \mu \end{aligned}$$

Intuition: Formula guarded reduction generates all possible disjuncts *that matter*.

[Altenkirch, Dybjer, Hofmann, Scott]

$$\frac{\Gamma \vdash M : \{\tau \mid \psi \mapsto t\} \quad \Gamma \vdash \psi}{\Gamma \vdash \text{out}_{\psi}^t M \equiv t : \tau}$$

$$\frac{\Gamma \vdash \alpha_i}{\Gamma \vdash \text{case}(\alpha_1.M_1, \alpha_2.M_2) \equiv M_i : \tau}$$

Updating Comparison at Neutral Types

$$\frac{M \rightarrow \overrightarrow{M}, \mu \quad \Gamma \vdash \bigvee_i \mu_i \quad \begin{array}{c} \tau \text{ is neutral} \\ \forall i. \Gamma, \mu_i \triangleright M_i \Leftrightarrow N : \tau \end{array}}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$

$$\frac{N \rightarrow \overrightarrow{N}, \nu \quad \Gamma \vdash \bigvee_i \nu_i \quad \begin{array}{c} \tau \text{ is neutral} \\ \forall i. \Gamma, \nu_i \triangleright M \Leftrightarrow N_i : \tau \end{array}}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$

$$\frac{\tau_1 \text{ is neutral} \quad \Gamma \triangleright M \Leftrightarrow N \uparrow \tau_1}{\Gamma \triangleright M \Leftrightarrow N : \tau_2}$$

Algorithmic Disjunction Extensionality

$$\frac{M \rightarrow \overrightarrow{M}, \mu \quad \Gamma \vdash \bigvee_i \mu_i \quad \begin{array}{c} \tau \text{ is neutral} \\ \forall i. \Gamma, \mu_i \triangleright M_i \Leftrightarrow N : \tau \end{array}}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$

Theorem

If $\Gamma \vdash \bigvee_i \alpha_i$ and $\Gamma, \alpha_i \triangleright M \Leftrightarrow N : \tau$, then $\Gamma \triangleright M \Leftrightarrow N : \tau$.

Algorithmic Disjunction Extensionality

Theorem

If $\Gamma \vdash \bigvee_i \alpha_i$ and $\Gamma, \alpha_i \triangleright M \Leftrightarrow N : \tau$, then $\Gamma \triangleright M \Leftrightarrow N : \tau$.

Proof.

Induction on the derivations of $\Gamma, \alpha_i \triangleright M \Leftrightarrow N : \tau$ with cases on τ .

- τ is a neutral type. Then,

$$\frac{M \rightarrow \overrightarrow{M}, \mu \quad \Gamma, \alpha_i \vdash \bigvee_i \mu_i \quad \begin{array}{c} \tau \text{ is neutral} \\ \forall i. \Gamma, \alpha_i, \mu_i \triangleright M_i \Leftrightarrow N : \tau \end{array}}{\Gamma, \alpha_i \triangleright M \Leftrightarrow N : \tau}$$

By induction on each subderivation, $\Gamma, \mu_i \triangleright M_i \Leftrightarrow N : \tau$. Since $\Gamma, \alpha_i \vdash \bigvee_j \mu_j$ for each i , we have $\Gamma, \bigvee_i \alpha_i \vdash \bigvee_j \mu_j$. Since $\Gamma \vdash \bigvee_i \alpha_i$, we have $\Gamma \vdash \bigvee_j \mu_j$. Then, $\Gamma \triangleright M \Leftrightarrow N : \tau$.



Algorithmic Disjunction Extensionality

Theorem

If $\Gamma \vdash \bigvee_i \alpha_i$ and $\Gamma, \alpha_i \triangleright M \Leftrightarrow N : \tau$, then $\Gamma \triangleright M \Leftrightarrow N : \tau$.

Proof.

Induction on the derivations of $\Gamma, \alpha_i \triangleright M \Leftrightarrow N : \tau$ with cases on τ .

- $\tau = (x : \tau_1) \rightarrow \tau_2(x)$. Then,

$$\frac{\Gamma, \alpha_i, x : \tau_1 \triangleright M \ x \Leftrightarrow N \ x : \tau_2(x)}{\Gamma, \alpha_i \triangleright M \Leftrightarrow N : (x : \tau_1) \rightarrow \tau_2(x)}$$

By induction on the subderivation, $\Gamma, x : \tau_1 \triangleright M \ x \Leftrightarrow N \ x : \tau_2(x)$.
Then, $\Gamma \triangleright M \Leftrightarrow N : (x : \tau_1) \rightarrow \tau_2(x)$.



Theorem

If $\Gamma \vdash \alpha : \mathcal{F}$ and $\Gamma \triangleright M \Leftrightarrow N : \tau$, then $\Gamma, \alpha \triangleright M \Leftrightarrow N : \tau$

Proof.

Induction on the derivation of $\Gamma \triangleright M \Leftrightarrow N : \tau$ with cases on τ .

- τ is a neutral type.

$$\frac{M \rightarrow \overrightarrow{M}, \mu \quad \Gamma \vdash \bigvee_i \mu_i \quad \begin{array}{c} \tau \text{ is neutral} \\ \forall i. \Gamma, \mu_i \triangleright M_i \Leftrightarrow N : \tau \end{array}}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$

By induction on the subderivation, $\Gamma, \mu_i, \alpha \triangleright M_i \Leftrightarrow N : \tau$. By weakening, $\Gamma, \alpha \vdash \bigvee_i \mu_i$. Thus, $\Gamma, \alpha \triangleright M \Leftrightarrow N : \tau$.



Definition expansion for extension types?

$$\frac{\begin{array}{l} \text{out}_{\psi}^t(M) \rightarrow (t, \psi), \overrightarrow{\text{out}_{\psi}^t(M), \mu} \quad \Gamma \vdash \bigvee_i \mu_i \vee \psi \\ \Gamma, \psi \triangleright t \Leftrightarrow t : \tau \\ \forall i. \Gamma, \mu_i \triangleright \text{out}_{\psi}^t(M_i) \Leftrightarrow t : \tau \end{array}}{\Gamma \triangleright \text{out}_{\psi}^t(M) \Leftrightarrow t : \tau}$$

Definition expansion for extension types?

$$\begin{array}{c}
 \text{out}_{\psi}^t(M) \rightarrow (t, \psi), \overrightarrow{\text{out}_{\psi}^t(M), \mu} \quad \begin{array}{l} \Gamma \vdash \bigvee_i \mu_i \vee \psi \\ \Gamma, \psi \triangleright t \Leftrightarrow t : \tau \\ \forall i. \Gamma, \mu_i \triangleright \text{out}_{\psi}^t(M_i) \Leftrightarrow t : \tau \end{array} \\
 \hline
 \Gamma \triangleright \text{out}_{\psi}^t(M) \Leftrightarrow t : \tau \\
 \\
 \begin{array}{l} \Gamma \vdash M : \tau \quad \Gamma, \psi \vdash t : \tau \quad \Gamma, \psi \vdash M \equiv t : \tau \\
 \hline
 \Gamma \vdash \text{in}_{\psi}^t M : \{\tau \mid \psi \mapsto t\} \end{array}
 \end{array}$$

β for disjunction eliminations?

$$\frac{\text{case}(\alpha_1.M_1, \alpha_2.M_2) \rightarrow (M_1, \alpha_1), (M_2, \alpha_2) \quad \begin{array}{l} \alpha_1, \alpha_1 \triangleright M_1 \Leftrightarrow M_1 : \tau \\ \alpha_1, \alpha_2 \triangleright M_1 \Leftrightarrow M_2 : \tau \end{array}}{\alpha_1 \triangleright \text{case}(\alpha_1.M_1, \alpha_2.M_2) \Leftrightarrow M_1 : \tau}$$

β for disjunction eliminations?

$$\frac{\text{case}(\alpha_1.M_1, \alpha_2.M_2) \rightarrow (M_1, \alpha_1), (M_2, \alpha_2) \quad \begin{array}{l} \alpha_1, \alpha_1 \triangleright M_1 \Leftrightarrow M_1 : \tau \\ \alpha_1, \alpha_2 \triangleright M_1 \Leftrightarrow M_2 : \tau \end{array} \quad \alpha_1 \vdash \alpha_1 \vee \alpha_2}{\alpha_1 \triangleright \text{case}(\alpha_1.M_1, \alpha_2.M_2) \Leftrightarrow M_1 : \tau}$$

$$\frac{\Gamma \vdash \alpha_1 \vee \alpha_2 \quad \Gamma, \alpha_1 \vdash M_1 : \tau \quad \Gamma, \alpha_2 \vdash M_2 : \tau \quad \Gamma, \alpha_1, \alpha_2 \vdash M_1 \equiv M_2 : \tau}{\Gamma \vdash \text{case}(\alpha_1.M_1, \alpha_2.M_2) : \tau}$$

Formula Guarded Reduction 2.0 (digression)

$$\begin{aligned}\Gamma \triangleright \text{case}(\mu_1.M_1, \mu_2.M_2) &\rightarrow (M_i, \top) && \text{if } \Gamma \vdash \mu_i \\ \Gamma \triangleright \text{case}(\mu_1.M_1, \mu_2.M_2) &\rightarrow (M, \mu_1), (M, \mu_2) \\ \Gamma \triangleright \text{out}_{\psi}^t(M) &\rightarrow (t, \top) && \text{if } \Gamma \vdash \psi \\ \Gamma \triangleright (\lambda x : \tau. M)N &\rightarrow (M[N/x], \top)\end{aligned}$$

Theorem

If $\Gamma \vdash \alpha : \mathcal{F}$ and $\Gamma \triangleright M \Leftrightarrow N : \tau$, then $\Gamma, \alpha \triangleright M \Leftrightarrow N : \tau$

Proof.

$$\frac{\Gamma \triangleright M \rightarrow \overrightarrow{M, \mu} \quad \Gamma \vdash \bigvee_i \mu_i \quad \forall i. \Gamma, \mu_i \triangleright M_i \Leftrightarrow N : \tau}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$



Attempt at Formula Weakening (digression)

Theorem

If $\Gamma \vdash \alpha : \mathcal{F}$ and $\Gamma \triangleright M \Leftrightarrow N : \tau$, then $\Gamma, \alpha \triangleright M \Leftrightarrow N : \tau$

Proof.

$$\frac{\Gamma \triangleright M \rightarrow \overrightarrow{M, \mu} \quad \Gamma \vdash \bigvee_i \mu_i \quad \forall i. \Gamma, \mu_i \triangleright M_i \Leftrightarrow N : \tau}{\Gamma \triangleright M \Leftrightarrow N : \tau}$$

Reduction is sensitive to formula weakening: $\Gamma, \alpha \triangleright M \rightarrow \overrightarrow{M', \mu'}$. We could try to prove that $\Gamma, \alpha, \mu_i \triangleright M'_i \Leftrightarrow M_i : \tau$, and use transitivity to get $\Gamma, \alpha, \mu_i \triangleright M'_i \Leftrightarrow N : \tau$. □

Attempt at Transitivity (digression)

Theorem

If $\Gamma \triangleright M \Leftrightarrow N : \tau$ and $\Gamma \triangleright N \Leftrightarrow O : \tau'$, then $\Gamma \triangleright M \Leftrightarrow O : \tau$.

Proof.

$$\frac{\Gamma \triangleright M \rightarrow \overrightarrow{M}, \mu_i \quad \Gamma \vdash \bigvee_i \mu_i \quad \forall i. \Gamma, \mu_i \triangleright M_i \Leftrightarrow O : \tau}{\Gamma \triangleright M \Leftrightarrow O : \tau} \quad \Gamma \triangleright O \Leftrightarrow N : \tau$$

Need $\Gamma, \mu_i \triangleright O \Leftrightarrow N : \tau$ to apply the induction hypothesis. □

Symmetry and Transitivity

$$\begin{aligned} & \text{case}(\mu_1.M_1, \mu_2.M_2) \rightarrow (M, \mu_1), (M, \mu_2) \\ \text{out}_{\psi}^t(M) & \rightarrow (t, \psi), (\text{out}_{\psi}^t(M_1), \mu_1), \dots, (\text{out}_{\psi}^t(M_l), \mu_l) \quad \text{if } M \rightarrow \overrightarrow{M}, \mu \end{aligned}$$

Theorem

1. If $\Gamma \triangleright M \Leftrightarrow N : \tau$ and $\Gamma \triangleright N \Leftrightarrow O : \tau$, then $\Gamma \triangleright M \Leftrightarrow O : \tau$.
2. If $\Gamma \triangleright M \Leftrightarrow N : \tau$, then $\Gamma \triangleright N \Leftrightarrow M : \tau$.

Lemma (Type Conversion)

1. If $\Gamma \triangleright M \Leftrightarrow N : \tau$, $\Gamma \vdash \tau \equiv \tau'$, then $\Gamma \triangleright M \Leftrightarrow N : \tau'$.
2. If $\Gamma \triangleright p \Leftrightarrow q \uparrow \tau$, then $\Gamma \triangleright p \Leftrightarrow q \uparrow \tau'$ with $\Gamma \vdash \tau \equiv \tau'$.

[Abel, Scherer], [Abel, Coquand, Manna]

$$C ::= [] \mid MC \mid \text{out}_{\psi}^t(C)$$

Theorem

$$\Gamma \triangleright C[\text{case}(\alpha_1.M_1, \alpha_2.M_2)] \Leftrightarrow \text{case}(\alpha_1.C[M_1], \alpha_2.C[M_2]) : \tau$$

Proof.

Formula guarded reduction reduces both sides to $(C[M_1], \alpha_1), (C[M_2], \alpha_2)$. For example:
 $\text{case}(\alpha_1.M_1, \alpha_2.M_2) \ N \downarrow (M_1 \ N, \alpha_1), (M_2 \ N, \alpha_2)$. □

Theorem

Let $\Gamma \vdash M : \tau$ and $\Gamma \vdash N : \tau$. If $\Gamma \triangleright M \Leftrightarrow N : \tau$, then $\Gamma \vdash M \equiv N : \tau$.

Lemma (Subject Reduction)

If $\Gamma \vdash M : \tau$ and $\Gamma \vdash M \rightarrow \overrightarrow{M}, \mu$, then, for all i , $\Gamma, \mu_i \vdash M \equiv M_i : \tau$ and $\Gamma, \mu_i \vdash M_i : \tau$.

Logical relation in the style of [Abel, Öhman, Vezzosi], [Abel, Coquand, Manna].

Theorem

If $\Gamma \vdash M \equiv N : \tau$, then $\Gamma \triangleright M \Leftrightarrow N : \tau$.

Second logical relation.

$$\frac{(\forall \Delta \leq \Gamma)(\Delta \Vdash M : \tau_1 \implies \Delta \Vdash \tau_2(M/x)) \quad (\forall \Delta \leq \Gamma)(\Delta \Vdash M = N : \tau_1 \implies \Delta \Vdash \tau_2(M/x) = \tau_2(N/x))}{\Gamma \Vdash (x : \tau_1) \rightarrow \tau_2(x)} \quad \Pi$$
$$\frac{\Gamma \vdash \psi : \mathcal{F} \quad \Gamma \Vdash \tau_1 \quad \Gamma, \psi \Vdash t : \tau_1}{\Gamma \Vdash \tau_1[\psi \rightarrow t]} \quad \text{Ext}$$

- If $\Gamma \Vdash \tau$ by Π then
 - $\Gamma \Vdash M = N : (x : \tau_1) \rightarrow \tau_2(x)$ iff
 $\Gamma \Vdash M : (x : \tau_1) \rightarrow \tau_2(x)$, $\Gamma \Vdash N : (x : \tau_1) \rightarrow \tau_2(x)$, and
 $\Gamma \vdash M \equiv N : (x : \tau_1) \rightarrow \tau_2(x)$, $\Gamma \vdash M \downarrow : M' : (x : \tau_1) \rightarrow \tau_2(x)$,
 $\Gamma \vdash N \downarrow : N' : (x : \tau_1) \rightarrow \tau_2(x)$ and for all $\Delta \leq \Gamma$,
 $\Delta \Vdash v : \tau_1 \implies \Delta \Vdash M'v = N'v : \tau_2(w/x)$.
- If $\Gamma \Vdash \tau$ by Ext then
 - $\Gamma \Vdash M = N : \tau[\psi \rightarrow t]$ iff
 $\Gamma \vdash M \downarrow : M' : \tau[\psi \rightarrow t]$, $\Gamma \vdash N \downarrow : N' : \tau[\psi \rightarrow t]$, $\Gamma \Vdash M : \tau[\psi \rightarrow t]$,
 $\Gamma \Vdash N : \tau[\psi \rightarrow t]$, $\Gamma \Vdash \text{out}_{\psi}^t(M') = \text{out}_{\psi}^t(N') : \tau$ and
 $\Gamma \vdash M \equiv N : \tau[\psi \rightarrow t]$.

- Proving completeness of optimized formula guarded reduction and adding in universes (preliminary version available in my master's thesis).
- Robust implementation of $\lambda^{\mathcal{L}}$ that supports user-defined logics.
- Extending this work to coproducts.

Thank you! Questions?

$$\frac{\Gamma \vdash p : \tau_1 + \tau_2 \quad \begin{array}{l} \Gamma, z : \tau_1 \vdash M[\iota_1(z)/x] \equiv N[\iota_1(z)/x] : \tau \\ \Gamma, z : \tau_2 \vdash M[\iota_2(z)/x] \equiv N[\iota_2(z)/x] : \tau \end{array}}{\Gamma \vdash M[p/x] \equiv N[p/x] : \tau}$$

$$\delta(p)(x_1 : \tau_1.N_1, x_2 : \tau_2.N_2) \downarrow \\ (N_1[z/x], p :=^{z:\tau_1} \iota_1(z) : \tau_1 + \tau_2), (N_2[z/x], p :=^{z:\tau_2} \iota_2(z) : \tau_1 + \tau_2)$$

$$\text{comp} : (M : \tau_i[]) (N : \tau[(i = 0) \rightarrow M @ O]) \rightarrow \tau_i[M @ 0, N @ 1]$$

$$\text{path}^{i,j} : \tau \left[\begin{array}{l} (i = 0) \mapsto p_2 @ j \\ (i = 1) \mapsto p_3 @ j \\ (j = 0) \mapsto p_0 @ x \\ (j = 1) \mapsto p_1 @ x \end{array} \right]$$

[Zhang]

$$\text{comp}^j(\tau[(i = 0) \vee (i = 1)] \mapsto [(i = 0) \mapsto M_0, (i = 1) \mapsto Q@j])(P@i)$$

- Formula: $(i = 0) \vee (i = 1)$
- Type of functions out of formulas: $(i = 0) \vee (i = 1) \rightarrow \tau$
- Type of terms of τ defined on the extent of $(i = 0) \vee (i = 1)$:
 $\tau[(i = 0) \vee (i = 1) \mapsto t]$

